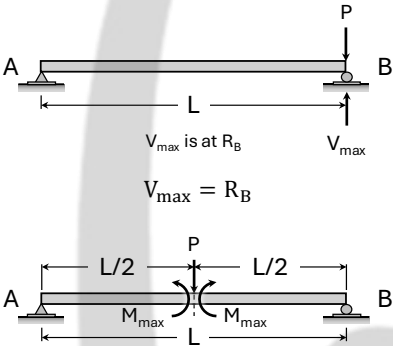
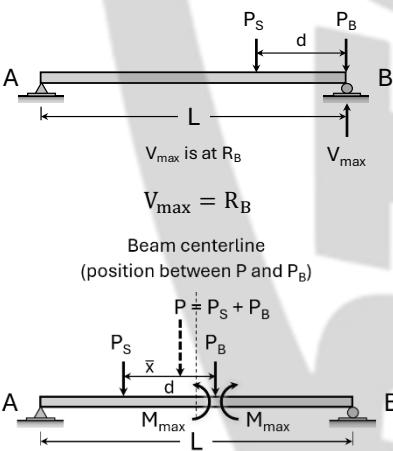
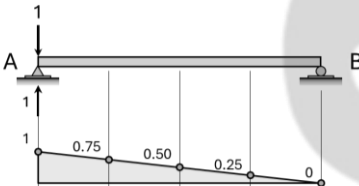
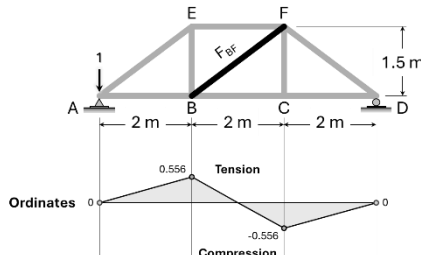


Influence Lines for Moving Loads Lecture Notes

Notes	Formulas
Moving Loads	A moving load is a load that travels along a structure, changing its position with time, such as vehicles or trains crossing a bridge. One Moving Load (P) Position for Max Shear and Moment:  $V_{\max}$ is at R_B $V_{\max} = R_B$ $M_{\max}$ is at midspan $M_{\max} = M_{\text{mid}} = \frac{PL}{4}$
	Two Moving Loads (P_s and P_b) Position for Max Shear and Moment:  $V_{\max}$ is at R_B $V_{\max} = R_B$ Beam centerline (position between P and P_B) $P = P_s + P_b$ $M_{\max}$ is under the bigger load P_b $M_{\max} = M_{\text{under } P_b} = \frac{(PL - P_s d)^2}{4PL}$ Note: If $2d \leq L$, possible for P_b only inside span: $\rightarrow M_{\max} = P_b L/4$
	An influence line for a beam is a diagram that shows how a specific response at a particular point (reaction, shear, or bending moment) varies as a unit load moves across the span. It is used to find the maximum effect of moving loads.  Influence Line for Reaction at A
Influence Lines for Beams	

Influence Lines for Trusses	An influence line for a truss is a diagram that shows how the axial force in a specific member varies as a unit load moves across the span, applied at the bottom-chord joints. It is used to determine the maximum effect of moving loads on that member.
	 Influence Line for Axial Force in Diagonal Member BF

Influence Lines for Moving Loads Lecture Problems

Moving Loads

Situation 1 – CE Board A man weighing 1.2 kN walks on a 3 m simply supported wooden log having a diameter of 250 mm.

- Determine the maximum shear (kN).
A. 0.6 B. 0.8 **C. 1.2** D. 1.8
- Determine the maximum bending moment (kN-m).
A. 1.5 B. 0.8 C. 1.2 **D. 0.9**
- If the log is replaced with a rectangular wooden section 300 mm wide and 100 mm thick, determine the maximum allowable weight (kN) of the man if the allowable flexural stress is $F_b = 4.2$ MPa.
A. 4.1 B. 2.4 **C. 2.8** D. 3.2

Situation 2 – CE Board Steel beams are used to support the deck of a bridge with a simple span of 25 m. The live load on each beam consists of wheel loads as follows:

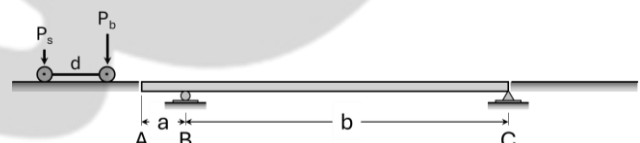
Front wheel load = 35.6 kN
Rear wheel load = 142.4 kN
Wheelbase = 4.3 m

- Determine the maximum support reaction (kN) in the beam.
A. 213 B. 268 **C. 172** D. 198
- Determine the maximum bending moment (kN-m) in the beam.
A. 1058 B. 1288 C. 1369 **D. 1037**
- Determine the maximum shear (kN) at the midspan of the beam.
A. 82.9 B. 96.5 C. 104.1 D. 128.5

Situation 3 – CE Board A truck crosses a bridge girder as shown in the figure.

Given: $a = 3$ m, $b = 22$ m, $P_s = 19.6$ kN, $P_b = 78.6$ kN, $d = 4.3$ m

- Determine the maximum positive bending moment (kN-m) that can be developed in the girder.
A. 572.3 B. 583.1 **C. 498.8** D. 656.6
- Determine the maximum negative bending moment (kN-m) that can be developed in the girder.
A. 58.8 B. 84.3 **C. 235.8** D. 338.0
- Determine the maximum negative shear (kN) that can be developed in the girder.
A. 78.6 **B. 94.4** C. 107.8 D. 45.3



Influence Lines

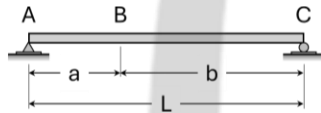
Situation 4 – CE Board Refer to the figure shown.

Given: $L = 9\text{ m}$, $a = 3\text{ m}$, $b = 6\text{ m}$

10. Determine the maximum ordinate of the influence line for the reaction at support A.
A. 2.00 B. 0.67 C. 1.00 D. 0.33

11. Determine the maximum ordinate of the influence line for the shear at point B.
A. 2.00 B. 0.67 C. 1.00 D. 0.33

12. Determine the maximum ordinate of the influence line for the bending moment at point B.
A. 2.00 B. 0.67 C. 1.00 D. 0.33



Situation 5 – Refer to the figure shown.

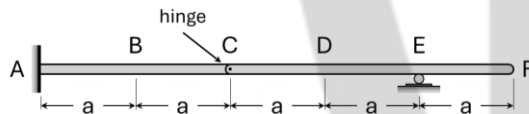
Given: $a = 1.5\text{ m}$

13. Determine the maximum ordinate of the influence line for the reaction at support A.
A. 0.50 B. 1.00 C. 0.75 D. 3.00

14. Determine the maximum ordinate of the influence line for the shear at point B.
A. 0.50 B. 1.00 C. 0.75 D. 3.00

15. Determine the maximum ordinate of the influence line for the reaction bending moment (kN-m) at support A.
A. 0.50 B. 1.00 C. 0.75 D. 3.00

16. Determine the maximum ordinate of the influence line for the bending moment (kN-m) at point D.
A. 0.50 B. 1.00 C. 0.75 D. 3.00



Situation 6 – CE Board Refer to the compound bridge girder shown.

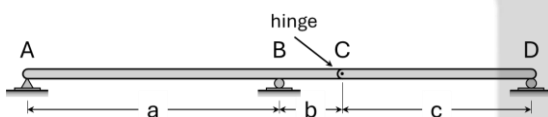
The girder will be subjected to a uniformly distributed live load.

Given: $a = 20\text{ m}$, $b = 5\text{ m}$, $c = 15\text{ m}$

17. For maximum reaction at support A, determine the length (m) of the girder that should be loaded with the uniformly distributed live load.
A. 10 B. 15 C. 20 D. 40

18. For maximum reaction at support D, determine the length (m) of the girder that should be loaded with the uniformly distributed live load.
A. 10 B. 15 C. 20 D. 40

19. For maximum reaction at support B, determine the length (m) of the girder that should be loaded with the uniformly distributed live load.
A. 10 B. 15 C. 20 D. 40



Situation 7 – CE Board Refer to the compound bridge girder shown.

Given: $a = 4\text{ m}$, $b = 10\text{ m}$

The girder will be subjected to the following loads:

Uniformly distributed dead load = 13 kN/m

Uniformly distributed live load = 12 kN/m

20. Determine the maximum reaction (kN) at support C.

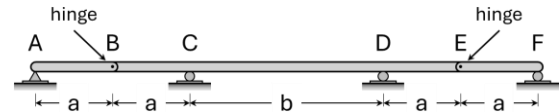
A. 294.2 B. 306.8 C. 275.0 D. 143.0

21. Determine the maximum shear (kN) at the midspan.

A. 37.1 B. 71.3 C. 34.2 D. 22.4

22. Determine the maximum bending moment (kN-m) at the midspan.

A. 192.0 B. 237.5 C. 150.0 D. 5.5



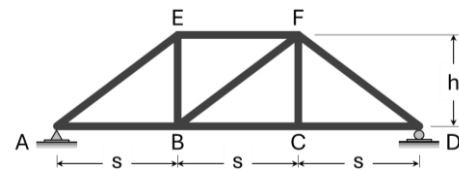
Situation 8 – The bridge truss shown in the figure will be subjected to moving loads.

Given: $s = 2\text{ m}$, $h = 1.5\text{ m}$

23. Determine the maximum ordinate of the influence line for the axial force in member EF.
A. 0.67 B. 0.56 C. 0.44 D. 0.89

24. Determine the maximum ordinate of the influence line for the axial force in member BC.
A. 0.67 B. 0.56 C. 0.44 D. 0.89

25. Determine the maximum ordinate of the influence line for the axial force in member BF.
A. 0.67 B. 0.56 C. 0.44 D. 0.89



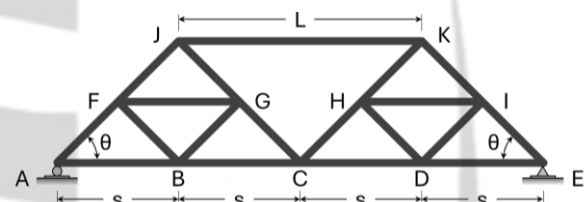
Situation 9 – CE Board The bridge truss shown in the figure will be subjected to moving loads.

Given: $s = 4\text{ m}$, $L = 8\text{ m}$, $\theta = 45^\circ$

26. Determine the maximum ordinate of the influence line for the axial force in member JK.
A. 1.50 B. 0.50 C. 0.75 D. 1.00

27. Determine the axial force (kN) in member JK due to a standard H-loading with a wheelbase of 4.3 m , front wheel load of 72.4 kN , and rear wheel load of 19.6 kN .
A. 63.25 C B. 63.25 T C. 81.47 C D. 81.47 T

28. Determine the axial force (kN) in member JK due to a uniformly distributed moving load of 9.35 kN/m .
A. 74.8 C B. 74.8 T C. 52.6 T D. 52.6 C



Situation 10 – CE Board The truss shown is to be analyzed by a lane load, truck load, and tandem load.

Given: $s = 4.300\text{ m}$, $h = 2.866\text{ m}$

Ordinates for member FG axial force influence line:

A = 0.000, B = 1.125, C = 0.750, D = 0.375, E = 0.000

Loads:

Lane load:

Uniformly distributed = 9.34 kN/m

Truck load:

Rear wheel = 145 kN , Front wheel = 35 kN , Wheelbase = 4.3 m

Tandem load:

Rear wheel = 108 kN , Front wheel = 108 kN , Wheelbase = 1.2 m

29. Determine the force (kN) in member FG due to the truck load.

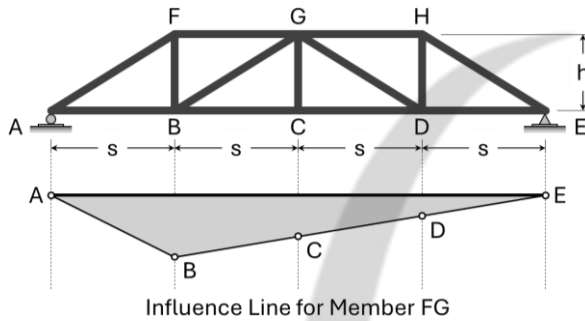
A. 271.9 B. 148.1 C. 189.4 D. 163.1

30. Determine the force (kN) in member FG due to the tandem load.

A. 202.5 B. 121.5 C. 209.1 D. 231.7

31. Determine the reaction (kN) at support E due to the lane load and truck load with 30% impact factor.

A. 302.9 B. 363.6 C. 336.6 D. 327.0



Influence Lines for Moving Loads Practice Problems

Moving Loads

Situation 11 – CE Board A truck with axle loads of 35 kN and 185 kN on a wheelbase of 4.3 m crosses across a 25-m simple span bridge.

32. Determine the maximum support reaction (kN).
A. 226 B. 214 C. 195 D. 238
33. Determine the maximum shear (kN).
A. 226 B. 214 C. 195 D. 238
34. Determine the maximum bending moment (kN-m).
A. 1300.8 B. 1350.5 C. 1421.7 D. 1294.2

Situation 12 – CE Board A girder of a bridge with a span of 24 m is simply supported at its ends. The girder is crossed by a standard H-loading with a wheelbase of 4.3 m, front wheel load of 71.2 kN, and rear wheel load of 17.8 kN.

35. Determine the maximum support reaction (kN).
A. 41.3 B. 96.3 C. 85.8 D. 102.4
36. Determine the maximum shear (kN) at midspan.
A. 41.3 B. 96.3 C. 85.8 D. 102.4
37. Determine the maximum bending moment (kN-m).
A. 632.8 B. 724.2 C. 496.4 D. 587.4

Situation 13 – CE Board The deck of a bridge consists of ribbed metal deck with 100 mm concrete slab on top. The superstructure supporting the deck is made of wide flange steel beams strengthened by cover plate 16 mm × 260 mm one at the top and one at the bottom and is spaced 1.2 m on centers. The beams are simply supported over a span of 25 m.

The loads on each beam are as follows:

Dead load = 12 kN/m (including beam and deck weight)

Wheel live loads:

Front wheel = 18 kN

Rear wheel = 72 kN

Wheel base = 4.3 m

Impact factor = $\frac{15}{L + 37} \leq 30\%$

Properties of W850×185:

$A = 23,750 \text{ mm}^2$

$t_w = 15 \text{ mm}$

$d = 850 \text{ mm}$

$I_x = 2662 \times 10^6 \text{ mm}^4$

$b_f = 290 \text{ mm}$

$I_y = 81.52 \times 10^6 \text{ mm}^4$

$t_r = 20 \text{ mm}$

38. Calculate the maximum bending stress (MPa) in the beam due to dead load.

- A. 123 B. 107 C. 92 D. 98

39. Calculate the maximum bending stress (MPa) in the beam due to live load plus impact.

- A. 79 B. 62 C. 68 D. 56

40. Calculate the maximum average web shear stress (MPa) in the beam due to live load plus impact.

- A. 7.6 B. 8.5 C. 9.1 D. 12.4

Influence Lines

Situation 14 – CE Board Refer to the figure shown.

Given: $L = 20 \text{ m}$, $a = 12 \text{ m}$, $b = 8 \text{ m}$

41. Determine the maximum ordinate of the influence line for the reaction at support A.

- A. 4.80 B. 0.60 C. 1.00 D. 0.40

42. Determine the maximum ordinate of the influence line for the shear at point B.

- A. 4.80 B. 0.60 C. 1.00 D. 0.40

43. Determine the maximum ordinate of the influence line for the bending moment (kN-m) at point B.

- A. 4.80 B. 0.60 C. 1.00 D. 0.40



Situation 15 – Refer to the figure shown.

Given: $a = 2 \text{ m}$

44. Determine the maximum ordinate of the influence line for the reaction at support A.

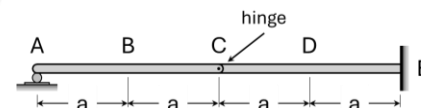
- A. 4.00 B. 2.00 C. 0.50 D. 1.00

45. Determine the maximum ordinate of the influence line for the shear at point B.

- A. 4.00 B. 2.00 C. 0.50 D. 1.00

46. Determine the maximum ordinate of the influence line for the bending moment (kN-m) at point E.

- A. 4.00 B. 2.00 C. 0.50 D. 1.00



Situation 16 – CE Board A simply supported beam has a span of 8 m with a 2 m overhang at the right end. The beam is subjected to a moving load and is to be analyzed for maximum shear at the midspan.

47. Determine the ordinate of the influence line for shear at the midspan of the beam.

- A. 0.30 B. 0.45 C. 0.25 D. 0.50

48. Determine the ordinate of the influence line for shear at the free end of the overhang.

- A. 0.30 B. 0.45 C. 0.25 D. 0.50

49. If a uniformly distributed moving load is applied on the beam. Determine the total length (m) of the beam that should be loaded to produce the maximum shear at the midspan.

- A. 4 B. 6 C. 3 D. 5

Situation 17 – CE Board An 8-m long beam is simply supported at the left end and at 2 m from the right end.

The girder will be subjected to the following loads:

Uniformly distributed live load = 9.35 kN/m

Concentrated live load, $P = 116 \text{ kN}$ for maximum shear

50. Determine the length (m) on the beam where the uniform load could act to produce maximum positive shear at midspan.

- A. 8 B. 2 C. 6 D. 4

51. Determine the length (m) on the beam where the uniform load could act to produce maximum negative shear at midspan.

- A. 2 B. 6 C. 4 D. 8

52. If the concentrated load is positioned at the end of the overhang, determine the maximum shear (kN) at the midspan.

- A. 67.4 B. 69.7 C. 38.4 D. 40.7

Situation 18 – CE Board A 25-m long girder is simply supported at the left end and at 3 m from the right end.

The girder will be subjected to the following loads:

Uniformly distributed live load = 9.35 kN/m

Concentrated live load, $P = 115 \text{ kN}$ for maximum shear

Concentrated live load, $P = 80$ kN for maximum moment

53. Determine the maximum shear (kN).

- A. 218 B. 268 C. 241 D. 365

54. Determine the maximum positive moment (kN-m).

- A. 1158 B. 1006 C. 1365 D. 854

55. Determine the maximum negative moment (kN-m).

- A. 312 B. 254 C. 296 D. 282

Situation 19 – CE Board Refer to the compound bridge girder shown.

Given: $a = 5$ m, $b = 20$ m, $c = 10$ m

56. For the maximum reaction at A, what length of the beam should the uniformly distributed load be applied?

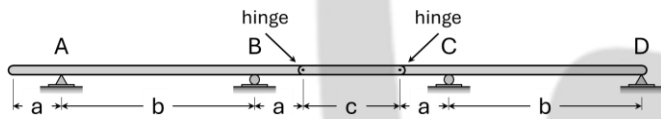
- A. 20 B. 25 C. 30 D. 45

57. For the maximum reaction at B, what length of the beam should the uniformly distributed load be applied?

- A. 5 B. 20 C. 40 D. 35

58. For the maximum reaction at C, what length of the beam should the uniformly distributed load be applied?

- A. 5 B. 20 C. 40 D. 35



Situation 20 – CE Board Refer to the compound bridge girder shown.

Given: $a = 5$ m, $b = 25$ m

The girder will be subjected to the following loads:

Uniformly distributed dead load, $w = 9.35$ kN/m

Concentrated live load, $P = 115$ kN for maximum reaction

Concentrated live load, $P = 80$ kN for maximum moment

59. For the maximum reaction at C, what length (m) of the beam should be applied with uniformly distributed load?

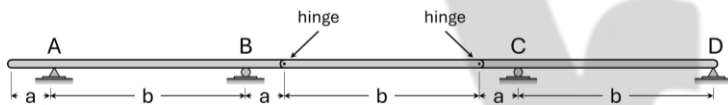
- A. 30 B. 50 C. 45 D. 55

60. What is the maximum reaction (kN) at A?

- A. 306.3 B. 278.3 C. 264.3 D. 420.3

61. For the maximum positive shear at midspan of AB, what length (m) of the beam should be applied with uniformly distributed load?

- A. 17.5 B. 60.0 C. 42.50 D. 50.5



Situation 21 – The bridge truss shown in the figure will be subjected to moving loads.

Given: $s = 4$ m, $h_1 = 3$ m, $h_2 = 1$ m

62. Determine the maximum ordinate of the influence line for the axial force in member FG.

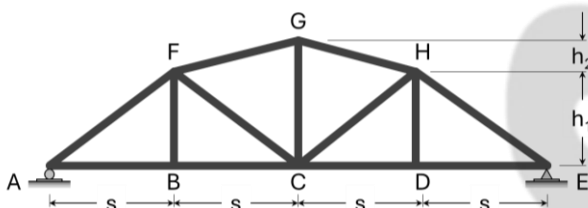
- A. 1.333 B. 0.667 C. 1.031 D. 0.515

63. Determine the axial force (kN) in member FG due to a standard H-loading with a wheelbase of 4.3 m, front wheel load of 72.4 kN, and rear wheel load of 19.6 kN.

- A. 84.0 B. 109.6 C. 108.6 D. 84.7

64. Determine the axial force (kN) in member FG due to a uniformly distributed moving load of 9.35 kN/m.

- A. 49.9 B. 99.7 C. 38.5 D. 77.1



Situation 22 – CE Board The bridge truss shown in the figure will be subjected to moving uniform highway load of 9.36 kN/m and a concentrated load of 37 kN.

Given: $s = 4$ m, $L = 8$ m

65. Determine the reaction (kN) at A due to the highway load alone.

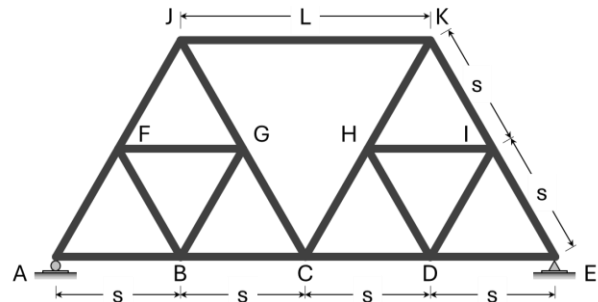
- A. 56.32 B. 93.38 C. 81.78 D. 74.88

66. Determine the maximum ordinate of the influence line for the axial force in member JK.

- A. 0.6325 B. 0.5774 C. 0.5000 D. 0.2887

67. Determine the axial force (kN) in member JK.

- A. 64.6 B. 96.9 C. 32.3 D. 129.2



Situation 23 – CE Board The figure shows a highway bridge truss.

Given: $s = 4$ m, $h = 5$ m

68. Find the maximum ordinate of the influence line for member FG.

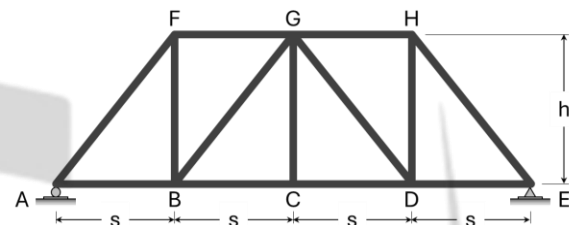
- A. 0.50 B. 1.0 C. 0.60 D. 0.75

69. Due to a uniformly distributed highway lane load, which length (m) of the bridge should be loaded for maximum compressive stress at member FG?

- A. 10 B. 14 C. 12 D. 16

70. For maximum stress at member FG, where should a concentrated highway load be located (m) from support at A?

- A. 0 B. 8.0 C. 6.0 D. 4.0



Situation 24 – CE Board It is required to compute the axial force in member FG of the bridge truss shown.

Given: $s = 5$ m, $h = 6$ m

Ordinates for member FG axial force influence line:

$A = 0.000$, $B = 0.625$, $C = 0.417$, $D = 0.208$, $E = 0.000$

71. Determine the maximum compressive force (kN) in member FG due to a uniform moving load of $w = 9.35$ kN/m.

- A. 93.5 B. 58.4 C. 43.8 D. 177.5

72. Determine the maximum compressive force (kN) in member FG due to two standard H-loading with front wheel load of 17.5 kN and rear wheel load of 71.2 kN which are 4.3 m apart.

- A. 44.5 B. 55.6 C. 51.9 D. 52.4

73. Determine the maximum compressive force (kN) in member FG due to a concentrated load of 116 kN and uniform load of $w = 9.35$ kN/m acting on span BC and CD only.

- A. 121.2 B. 111.4 C. 145.2 D. 130.9

